

Zaraius Bilimoria

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PROFESSIONAL SUMMARY

Robotics Engineering student at Olin College with hands-on autonomy stack experience spanning perception, sensor fusion, motion planning, and embedded systems. Built and deployed production robotics software for Boston Dynamics Spot and computer vision systems for real-world agricultural applications. Comfortable across the full stack from C++ ROS 2 nodes and custom Nav2 plugins to embedded hardware bring-up and ML inference pipelines.

EDUCATION

Olin College of Engineering (B.S. Robotics Engineering) • Needham, MA • GPA: 3.97

May 2028

SKILLS

Robotics & Embedded Systems: ROS 2, Nav2, Docker, Gazebo, RViz, Foxglove, Ubuntu, Raspberry Pi, STM32

Programming & Development: C++, Python, C, Java, MATLAB, Git, CMake, Bash, Tmux, Vim, Julia

Computer Vision & ML: PyTorch, OpenCV, YOLO, TensorRT, ZED SDK, Open3D, DeepStream SDK

Design & Fabrication: Onshape, SolidWorks, KiCad, 3D Printing, Soldering, Hand Tools, Power Tools

EXPERIENCE

Robotics Intern • HITT Contracting • Falls Church, VA

Jun 2026 – Aug 2026

- Architected a **ROS 2 social navigation** system for Boston Dynamics Spot, integrating human perception, prediction, and Nav2 motion planning for operation around pedestrians on active construction sites.
- Developed a **C++ ROS 2** perception pipeline using ZED 2i skeleton tracking to estimate pedestrian position, velocity, and posture classification from **38 body keypoints**, providing real-time human state information for navigation.
- Implemented a custom Nav2 MPPI SocialCritic plugin in C++ that predicts pedestrian trajectories and applies anisotropic Gaussian cost fields across **5,000 candidate trajectories** per planning cycle, enabling real-time socially-aware navigation around stationary and moving pedestrians.
- Validated the full autonomy stack in a live **on-site demo** on an active construction site, integrating perception, localization, and navigation and resolving TF, sensor synchronization, and coordinate-frame issues on Boston Dynamics Spot.
- Built a Docker-based **TurtleBot3** simulation environment for rapid development and repeatable validation of the autonomy stack before deployment to the physical robot.

Embedded Systems & Computer Vision Intern • Prophet AI • Somerville, MA

Jun 2025 – Dec 2025

- Designed and deployed a scalable **12-camera network** for a 100-bird poultry farm using Raspberry Pis, replacing a legacy IP-camera system. Programmed H.264-over-RTSP stream in Python, ensuring 24/7 uptime with automated IP provisioning.
- Boosted inference throughput from **220 to 3,000 FPS** and cut compute load by 75% by training a lightweight PyTorch YOLO model and building a custom pixel-based motion filter with OpenCV, allowing continuous poultry health monitoring.
- Increased the model F1 score from **0.94 to 0.99** by implementing a dynamic masking system to improve data quality, and by developing an active-learning pipeline which trains two CV models nightly, computes IoU disagreement, and flags discrepancies for retraining on flagged images.

Software & Integration Lead • Olin AERO • Needham, MA

Aug 2024 – Current

- Taught core robotics software concepts (ROS 2, state machines, computer vision, and networking) to 6 first-year team members, structuring mini-lectures and projects to ensure contributions to the SUAS competition aircraft.
- Developed and led the implementation of the communication system for an autonomous aerial vehicle using TCP to reliably transmit live images and flight data from onboard Pixhawk autopilot and Jetson Nano to a ground station.
- Integrated **Pixhawk autopilot** for bidirectional control and data flow with onboard computing systems.

Robotics Software Engineer • Olin Agriculture Robotics Research Team • Needham, MA

Aug 2024 – Current

- Designed and implemented ROS 2 perception and **sensor fusion** algorithms for real-time crop-row detection and global pose estimation, enabling autonomous navigation and alignment with crop rows in unstructured environments.
- Implemented **3D Gaussian splatting** techniques to process depth camera data, generating dense 3D point clouds for weed identification and path planning.
- Designed schematic and fabricated a protoboard-soldered signal-conditioning interface for commercial resistive soil probe.

Captain, Lead Programmer; NUTRONS 125 • FIRST Robotics • Revere, MA

Jan 2019 – Jul 2024

- Led a 19-member team to create a robot ranked **2nd out of 600+** teams at the global championship.
- Developed autonomous and teleoperated robot code in Java, collaborating with mentors and the mechanical team.